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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
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Dear Prof. Merz,

We submit our revised manuscript, **"Computational Discovery of Antimalarial Candidates from African Natural Product Chemical Space"**, for consideration as an Article in *JCIM* (Manuscript ID: ci-2026-012015). The work presents a resource-efficient, reproducible computational workflow for exploring African natural-product-inspired antimalarial chemical space, combining VAE-guided centroid selection, dual-filter docking, and multi-parameter optimization.

The principal contribution is a controlled reduction in screening cost: 484 chemically diverse centroids replace an estimated 65 856-molecule exhaustive screen, corresponding to a 99.3 % reduction in docking runs. The framework was evaluated using complementary, explicitly separated benchmarks. Prospective DEKOIS testing of single-method AutoDock Vina on PfDHFR (40 actives and 1200 property-matched decoys) gave near-random discrimination (ROC-AUC 0.450; 95 % CI 0.367–0.531). In contrast, the MMV Malaria Box analysis provides a positive-control, retrodictive consistency assessment for the Vina–DiffDock consensus (ROC-AUC 0.924–1.000 across the evaluated target panels); score-stratified target results are clearly distinguished from external decoy validation in the manuscript. These results motivate, but do not replace, prospective experimental testing.

Application to the hybrid library of 396 African natural products and 454 synthetic antimalarials identified 19 913 computationally prioritized leads predicted to be synthesizable. Structural analysis showed substantial molecular novelty (92.6 % of the evaluated sample below a maximum seed Tanimoto similarity of 0.4) alongside 69.3 % recovery of seed scaffolds, supporting a scaffold-preserving, substituent-diversifying search strategy. All activity, selectivity, ADMET, and binding results remain explicitly labelled as computational estimates. The manuscript also corrects the receptor identity associated with PDB 4GM2: it is reported as PfClpR-specific and is not presented as PfClpP validation.

We believe the manuscript fits *JCIM* because it addresses chemical-space exploration, molecular representation, virtual-screening benchmarking, multi-objective prioritization, and reproducible computational practice. Source code and the currently available data are versioned in the public repository (`https://github.com/NanaEngo/Malaria_codesV2`); an archival release will provide a citable dataset record.

This manuscript represents original work, is not under consideration elsewhere, and has been reviewed and approved by all authors. The authors declare no competing interests.

Sincerely,
Myke Vital Sao Temgoua
(on behalf of all co-authors)